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Practice Tuesday 11:00-12:00

Normalization

1. Consider the following unnormalized table used for printing booking receipts: `Booking_receipt(booking_id, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, seat_numbers, ticket_price)`. The `seat_numbers` column stores a comma-separated list (e.g. "12A, 12B") when a booking covers several passengers on the same reservation. Identify which normal form (1NF, 2NF, 3NF) this table violates and explain why, referring to the specific columns responsible.
2. Normalize `Booking_receipt` step by step: first bring it to 1NF, then to 2NF, then to 3NF. For each step draw/describe the resulting table(s), underline the primary key of each table, and mark the foreign keys.
3. List the functional dependencies you used to justify each normalization step in task 2 (e.g. `booking_id → flight_number`, `flight_number → airline_name`).
4. For the airport database you created in the DDL section, check whether the `Flights` table is in 3NF. State the current primary key and functional dependencies and explain whether any transitive dependency exists between non-key attributes (for example, via `departing_airport_id` or `airline_id`).
5. Using `CREATE TABLE` statements, implement the 3NF tables you designed in task 2, then write `INSERT` statements that populate them with at least 5 rows of sample data split out from a denormalized dataset of your choice.

DDL

1. Create following tables with corresponding attributes:
 - **Airline_info:** `airline_id (int), airline_code (varchar (30)), airline_name (varchar(50)), airline_country (varchar(50)), created_at (timestamp), updated_at (timestamp), info (varchar(50)) ;`
 - **Airport:** `airport_id (int), airport_name (varchar(50)), country (varchar(50)), state (varchar(50)), city (varchar(50)), created_at (timestamp), updated_at (timestamp) ;`
 - **Baggage_check:** `baggage_check_id (int), check_result (varchar(50)), created_at (timestamp), updated_at (timestamp), booking_id (int), passenger_id (int);`
 - **Baggage:** `baggage_id (int), weight_in_kg (decimal(4,2)), created_at (timestamp, updated_at (timestamp) , booking_id (int);`
 - **Boarding_pass:** `boarding_pass_id (int), booking_id (int), seat (varchar(50)), boarding_time (timestamp), created_at (timestamp), updated_at (timestamp);`

- **Booking_flight:** booking_flight_id(int), booking_id(int), flight_id(int), created_at(timestamp), updated_at(timestamp);
- **Booking:** booking_id(int), flight_id(int), passenger_id(int), booking_platform(varchar(50)), created_at(timestamp), updated_at(timestamp), status(varchar(50)), price(decimal(7,2));
- **Flights:** flight_id(int), sch_departure_time(timestamp), sch_arrival_time(timestamp), departing_airport_id(int), arriving_airport_id(int), departing_gate(varchar(50)), arriving_gate(varchar(50)), airline_id(int), act_departure_time(timestamp), act_arrival_time(timestamp), created_at(timestamp), updated_at(timestamp) ;
- **Passengers:** passenger_id(int), first_name(varchar(50)), last_name(varchar(50)), date_of_birth(date), gender(varchar(50)), country_of_citizenship(varchar(50)), country_of_residence(varchar(50)), passport_number(varchar(20)), created_at(timestamp), updated_at(timestamp) ;
- **Security_check:** security_check_id(int), check_result(varchar(20)), created_at(timestamp), updated_at(timestamp), passenger_id(int);

2. Define Primary Keys for each tables;

4. Define for all attributes not null constraint;

5. Rename airline_info table to airline;

6. Rename column price to ticket_price in booking table;

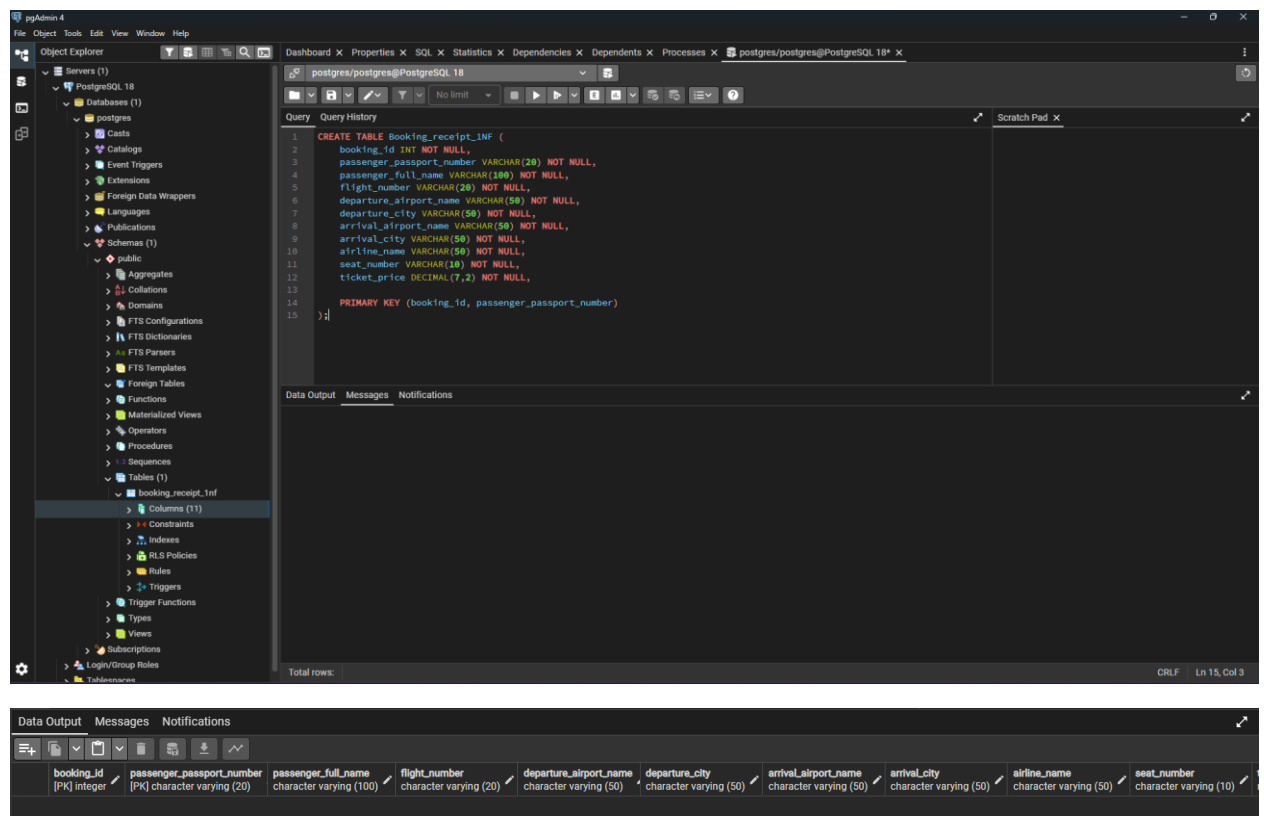
7. Change data type of departing_gate from varchar(50) to text;

8. Drop the column info(varchar(50)) from the airline table.

9. Make a relationship between following tables:

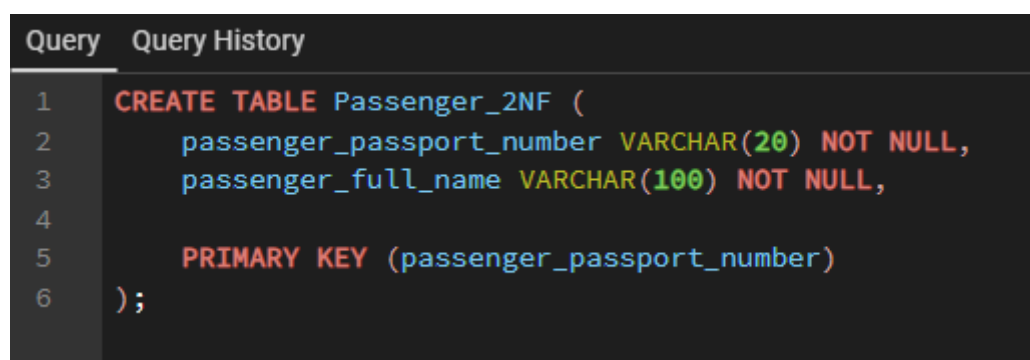
- Passengers with Secuitiry_check, Booking, Baggage_check by passenger_id;
- Booking with Baggage_check, Baggage, Boarding_pass, Booking_flight by booking_id;
- Flights with Booking_flight by flight_id;
- Airport with Flights by departing_airport_id;
- Airport with Flights by arriving_airport_id;
- Airline with Flights by airline_id;

First, to start normalization I created a table with all essential attributes and entities:



This is part of 1NF normalization, since I stored “12A and 12B” in a different column named “seat_number”

After the 1NF part completed, I removed partial dependencies, by separating columns into own tables, to fulfil 2NF conditions:



Query	Query History
1	CREATE TABLE Booking_2NF (
2	booking_id INT NOT NULL,
3	flight_number VARCHAR(20) NOT NULL,
4	ticket_price DECIMAL(7,2) NOT NULL,
5	
6	PRIMARY KEY (booking_id)
7	);

Query	Query History
1	CREATE TABLE Booking_Passenger_2NF (
2	booking_id INT NOT NULL,
3	passenger_passport_number VARCHAR(20) NOT NULL,
4	seat_number VARCHAR(10) NOT NULL,
5	
6	PRIMARY KEY (booking_id, passenger_passport_number),
7	
8	FOREIGN KEY (booking_id)
9	REFERENCES Booking_2NF(booking_id),
10	
11	FOREIGN KEY (passenger_passport_number)
12	REFERENCES Passenger_2NF(passenger_passport_number)
13	);

Here I made 3 new tables – “passenger”; “booking”, “booking_passenger”, assigned all PK’s and FK’s, and established relationships between them.

To start going from 2NF to 3NF, I removed more dependencies via separating flight, airport, airline into separate tables:

Airline:

Query	Query History
1	CREATE TABLE Airline_3NF (
2	airline_name VARCHAR(50) NOT NULL,
3	
4	PRIMARY KEY (airline_name)
5	);

Airport:

Query	Query History
1	CREATE TABLE Airport_3NF (
2	airport_name VARCHAR(50) NOT NULL,
3	city VARCHAR(50) NOT NULL,
4	
5	PRIMARY KEY (airport_name)
6	);

Flights:

Query	Query History
1	CREATE TABLE Flight_3NF (
2	flight_number VARCHAR(20) NOT NULL,
3	departure_airport_name VARCHAR(50) NOT NULL,
4	arrival_airport_name VARCHAR(50) NOT NULL,
5	airline_name VARCHAR(50) NOT NULL,
6	
7	PRIMARY KEY (flight_number),
8	
9	FOREIGN KEY (departure_airport_name)
10	REFERENCES Airport_3NF(airport_name),
11	
12	FOREIGN KEY (arrival_airport_name)
13	REFERENCES Airport_3NF(airport_name),
14	
15	FOREIGN KEY (airline_name)
16	REFERENCES Airline_3NF(airline_name)
17	);

I established relationships with “booking” and “flights” table:

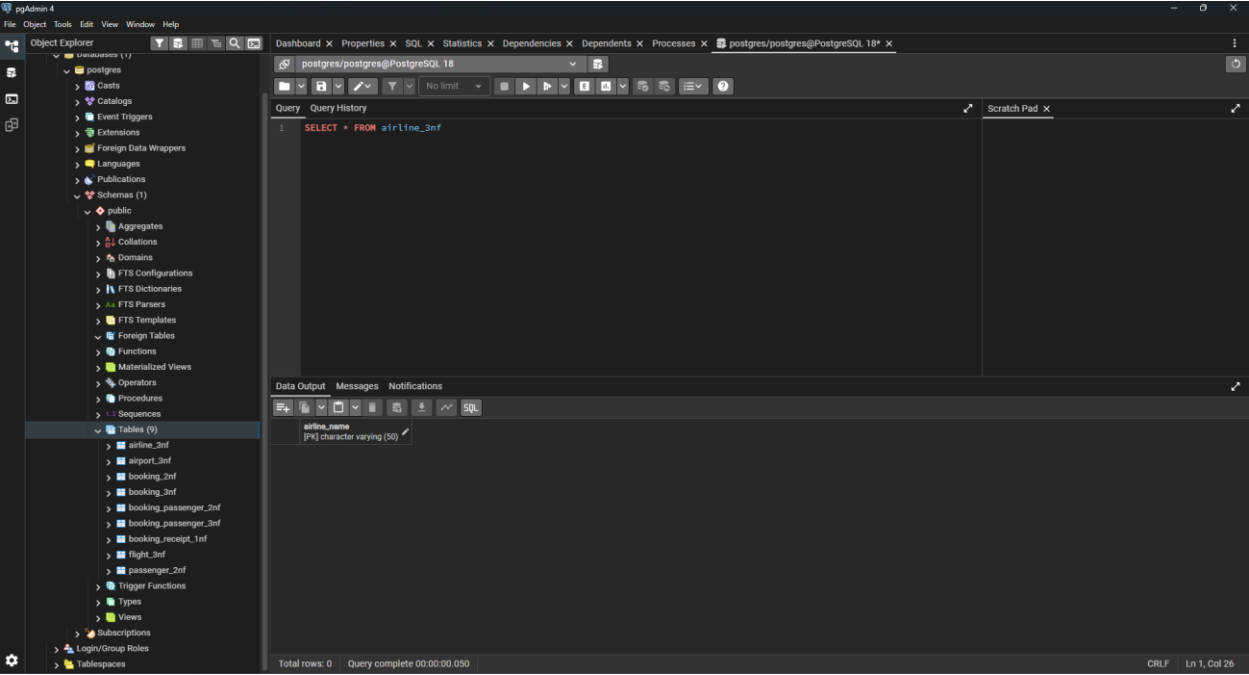
Query	Query History
1	CREATE TABLE Booking_3NF (
2	booking_id INT NOT NULL,
3	flight_number VARCHAR(20) NOT NULL,
4	ticket_price DECIMAL(7,2) NOT NULL,
5	
6	PRIMARY KEY (booking_id),
7	
8	FOREIGN KEY (flight_number)
9	REFERENCES Flight_3NF(flight_number)
10	);

And another relationship for “passengers” and “booking”:

Query	Query History
1	CREATE TABLE Booking_Passenger_3NF (
2	booking_id INT NOT NULL,
3	passenger_passport_number VARCHAR(20) NOT NULL,
4	seat_number VARCHAR(10) NOT NULL,
5	
6	PRIMARY KEY (booking_id, passenger_passport_number),
7	
8	FOREIGN KEY (booking_id)
9	REFERENCES Booking_3NF(booking_id),
10	
11	FOREIGN KEY (passenger_passport_number)
12	REFERENCES Passenger_2NF(passenger_passport_number)
13	);

Now we can see all the normalization steps and corresponding tables:

3NF:



airport_name [PK] character varying (50)	city character varying (50)
--	---------------------------------------

Data Output	Messages	Notifications
booking_id [PK] integer	flight_number character varying (20)	ticket_price numeric (7,2)

Data Output	Messages	Notifications
booking_id [PK] integer	passenger_passport_number [PK] character varying (20)	seat_number character varying (10)

Data Output

Messages

Notifications

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SQL

	flight_number [PK] character varying (20)	departure_airport_name character varying (50)	arrivalAirport_name character varying (50)	airline_name character varying (50)
--	--	--	---	--

2NF:

Data Output	Messages	Notifications
        		
	booking_id [PK] integer	flight_number character varying (20)
		ticket_price numeric (7,2)

Here, in the table “booking” according to assignment task, I renamed column price to “ticket_price”

Data Output	Messages	Notifications
        		
	booking_id [PK] integer	passenger_passport_number [PK] character varying (20)
		seat_number character varying (10)

Data Output	Messages	Notifications
        		
	passenger_passport_number [PK] character varying (20)	passenger_full_name character varying (100)

And 1NF table:

Data Output	Messages	Notifications
        		
	booking_id [PK] integer	passenger_passport_number [PK] character varying (20)
	passenger_full_name character varying (100)	flight_number character varying (20)
	departure_airport_name character varying (50)	departure_city character varying (50)
	arrival_airport_name character varying (50)	arrival_city character varying (50)
	airline_name character varying (50)	seat_number character varying (10)
1	101	P12345
2	101	P67890

Now, there are still some tasks remaining, first we need to change departing_gate from VARCHAR to TEXT:

Query	Query History
1	ALTER TABLE booking_receipt_1nf
2	ALTER COLUMN departing_gate TYPE TEXT;

And also drop “info” column from airline tables:

Query	Query History
1	ALTER TABLE booking_receipt_1nf
2	DROP COLUMN info;